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AI/ML for Nuclear Thermal Hydraulics

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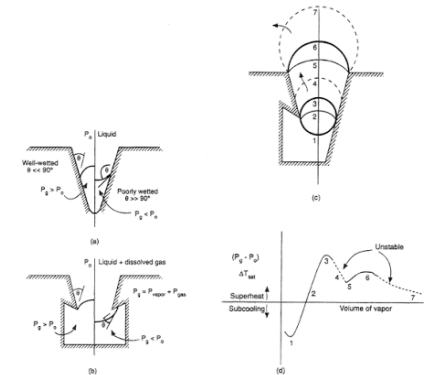
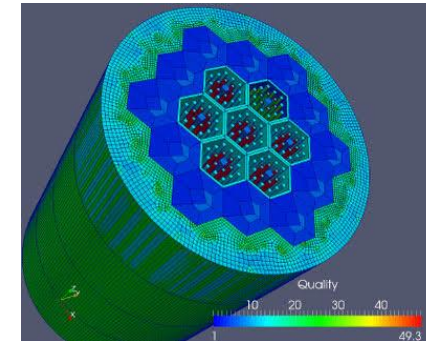
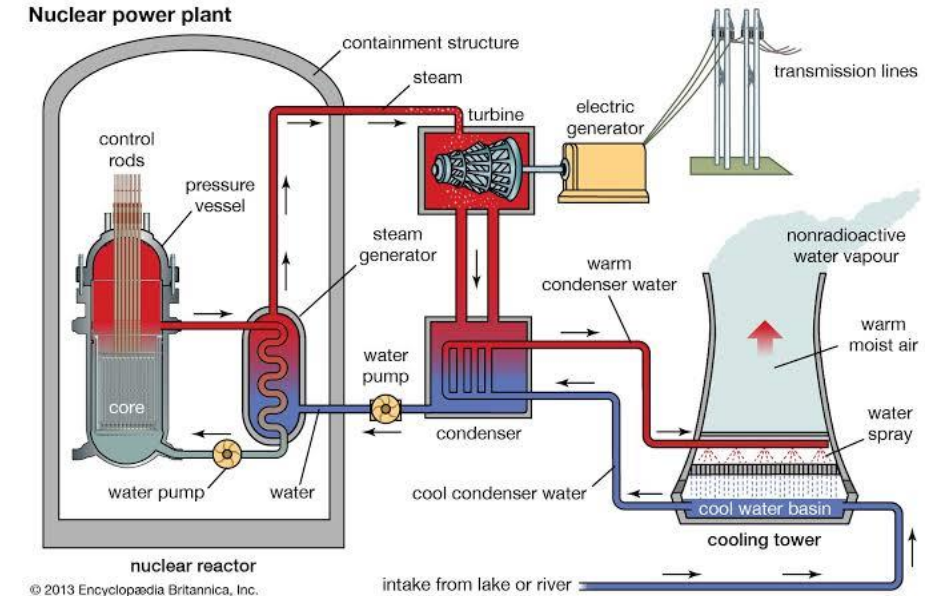
- Safety Requirements
- Thermal hydraulic challenges
- Simulation of TH systems
- Reactor analysis workflow
 - Pointwise, conservative, UQ aware
- Fast models
- AI/ML applications to NTH?
- Schedule

Safety Requirements

- All nuclear reactors rely on a working fluid (coolant) in order to transport energy from the core
- Unique to nuclear is
 - Its ability to operate at arbitrary power
 - Decay heat: significant heat (initially ~7% of operating generation) is still generated even after fission has ceased
- In all conceivable conditions, we must maintain a **margin to thermal limits** determined by the **structural integrity** of key components
- Conceivable conditions: determined by experience and available expert knowledge
- Inherently coupled to other physics (neutronics, structural)
- Inherently transient (& stiff)
- Inherently uncertain (aleatoric & epistemic)

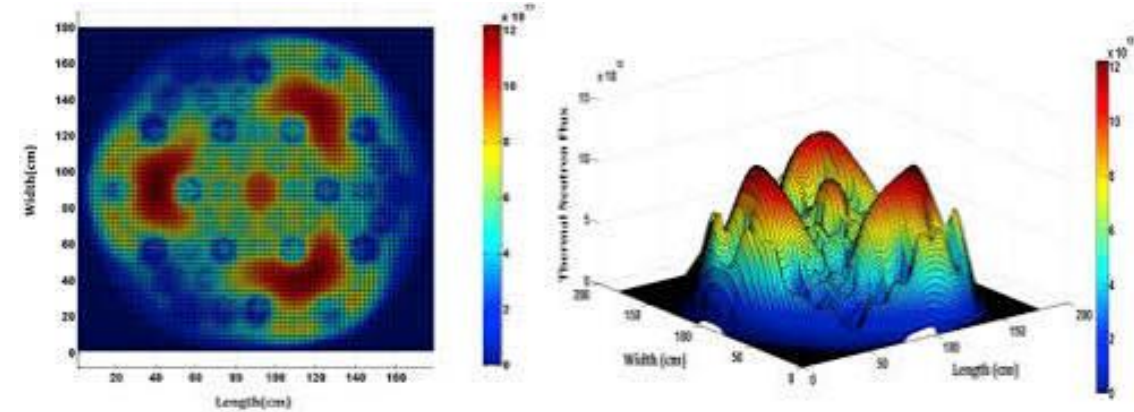
Thermal hydraulics challenges

- Nuclear thermal hydraulics is truly multiscale (in space and time)
- System-scale: heat exchange throughout entire primary circuit, and secondary circuit
- Core-scale: heat exchange from fuel pins to coolant in the core
- Component-scale: flows through key components (plena, pressurizer, T-junctions, steam generators, valves & pumps)
- Micro-scale: nucleate boiling, DNB, CHF



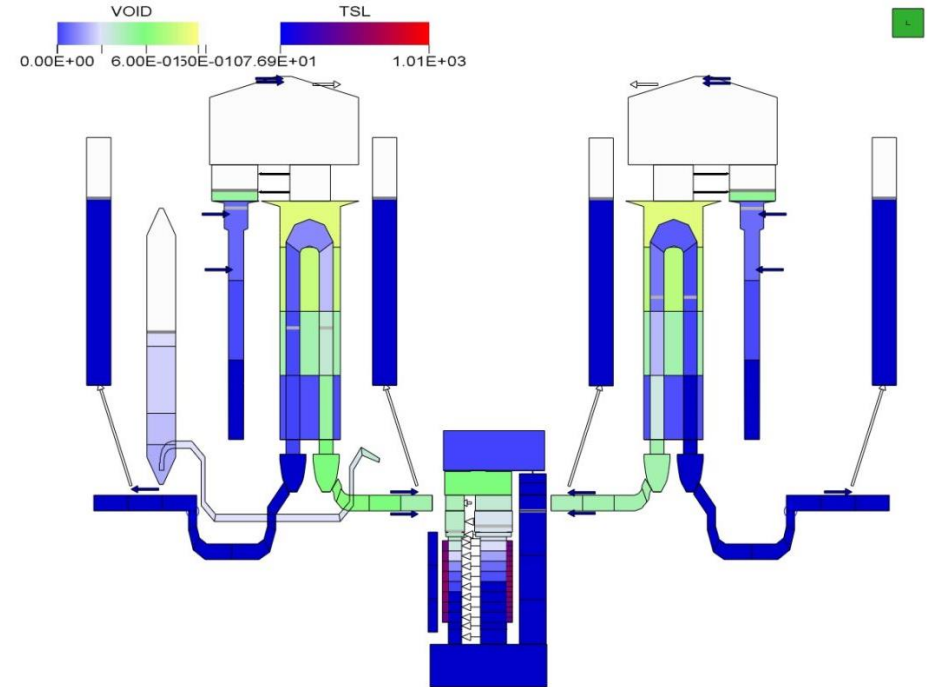
Thermal hydraulics challenges

- NTH is also a multiphysics problem
 - Coolant properties affect the neutronics
 - Neutronics determines local heat generation
 - Local heat generation determines coolant properties
 - and so on...



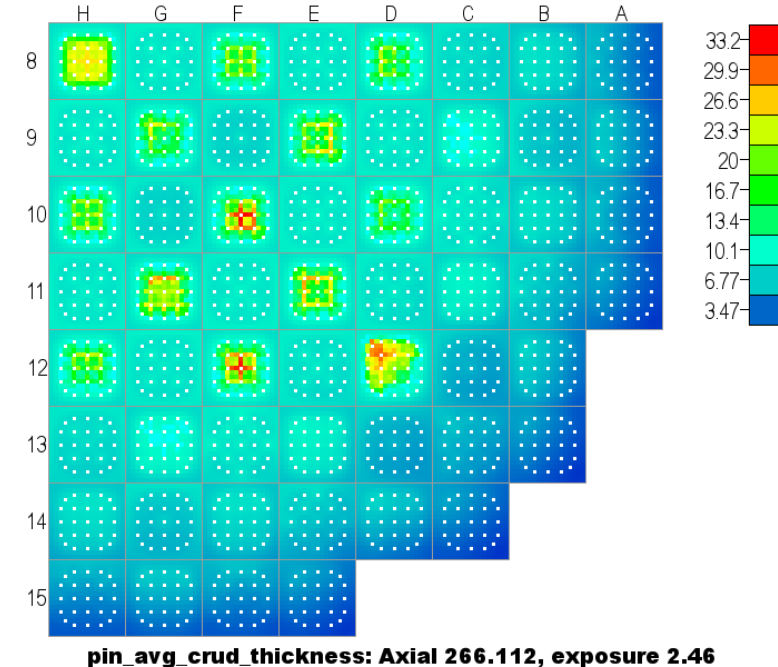
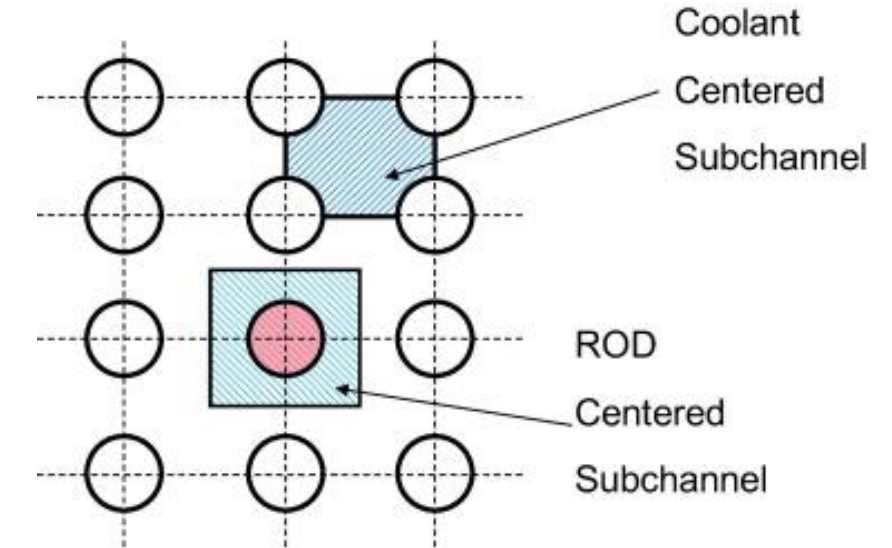
Simulation of TH systems: Systems codes

- System-scale: Systems codes
 - eg RELAP5-3D, ATHLET, CATHARE
- Simplified physics surrogate
 - Cross-section averaged channel flow
 - 1D 'nodalization'
- Flow regime map; relates flow state to wall/interface correlations
- 0D models for pumps, valves & trips
- Rapid (~real time)



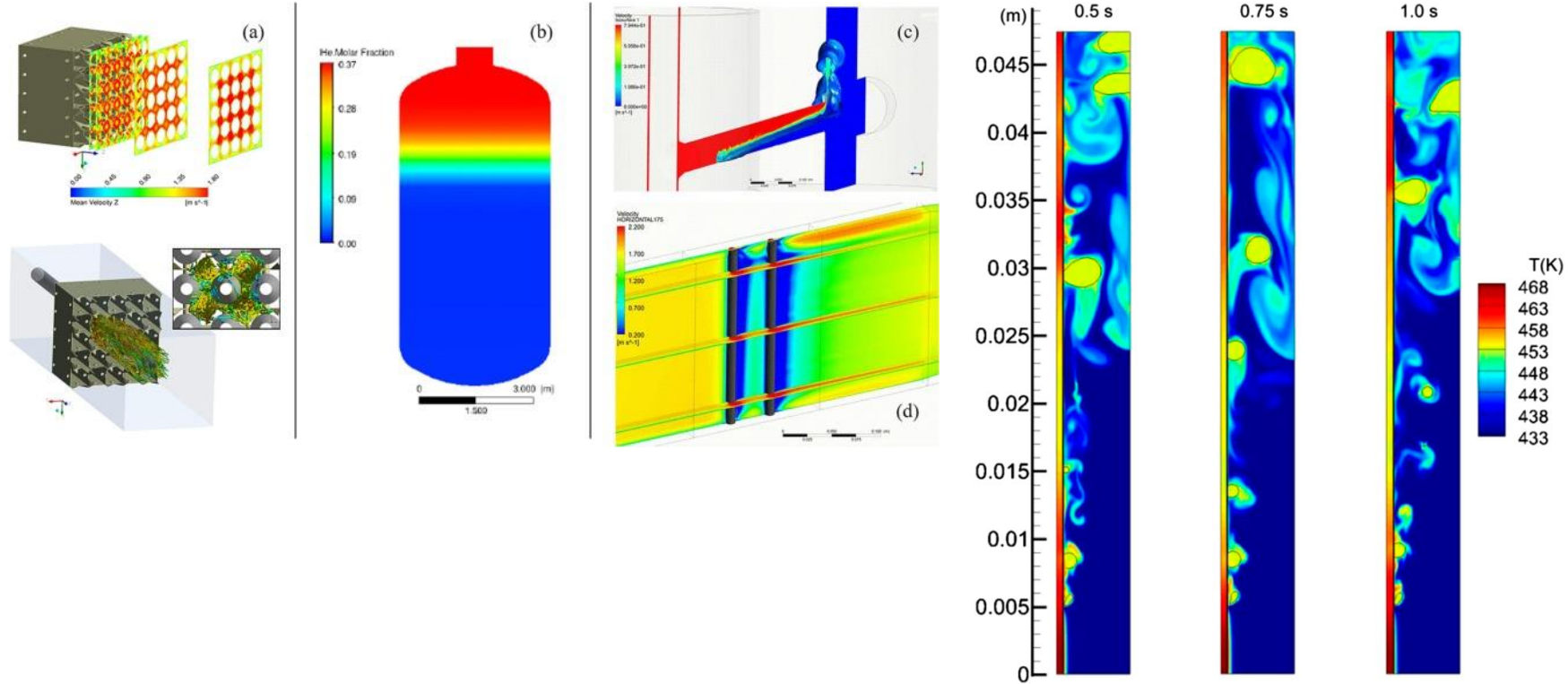
Simulation of TH systems: Subchannel codes

- Core-scale: Subchannel codes
 - eg CTF, VIPRE, SUBCHANFLOW
- Simplified physics surrogate
 - Quasi 3D
 - Provides for lateral cross flow
- Flow regime map; relates flow state to wall/interface correlations
- Subgrid CFD



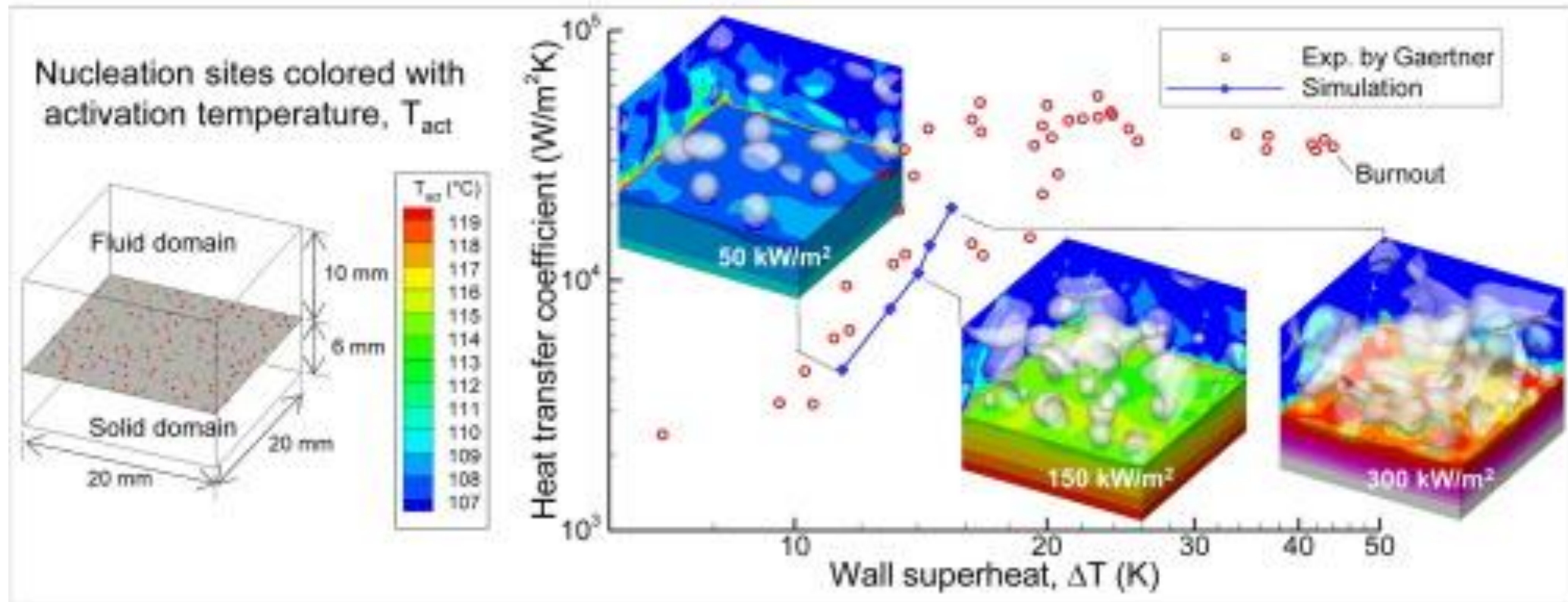
Simulation of TH systems: CFD

- Component-scale: Computational fluid dynamics (CFD)
- LES
- URANS
- RANS
- Multiphase
 - Eulerian-eulerian
 - Lagrangian
- Interface tracking



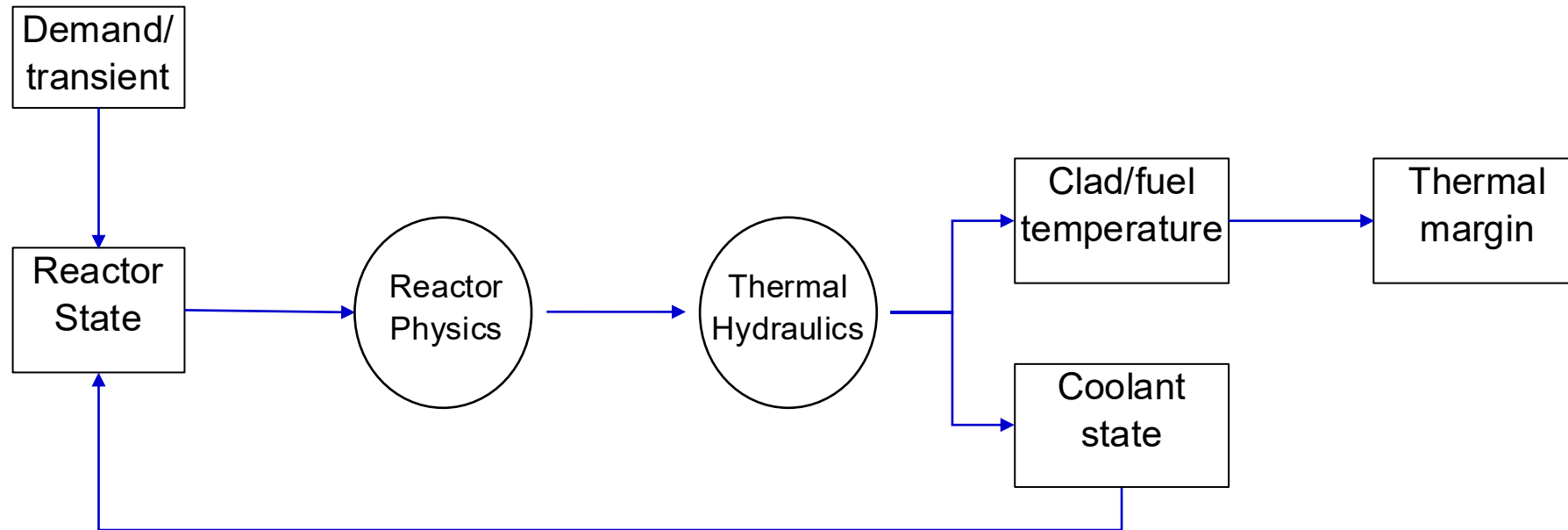
Simulation of TH systems: Interface tracking

- Micro-scale: CFD
- Interface tracking
- Molecular dynamics



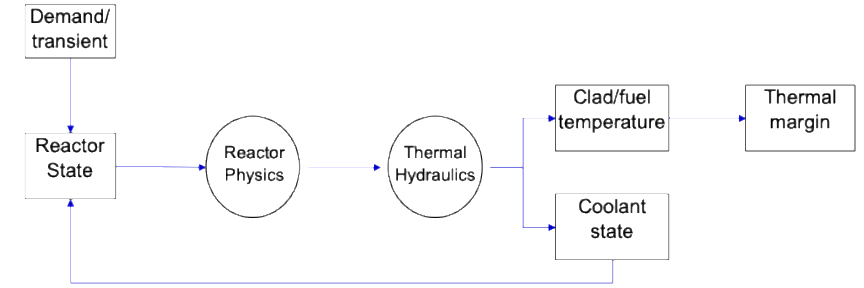
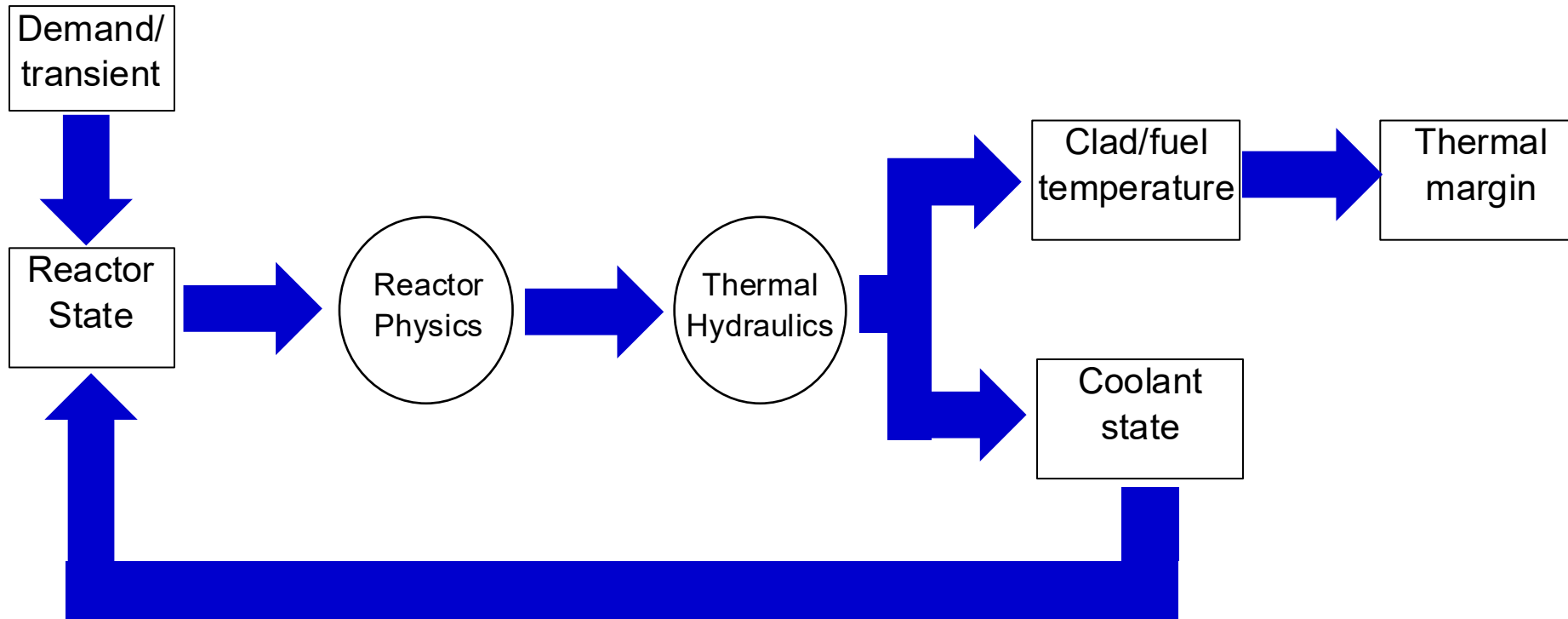
Reactor analysis workflow

- Point-wise analysis



Conservative workflow

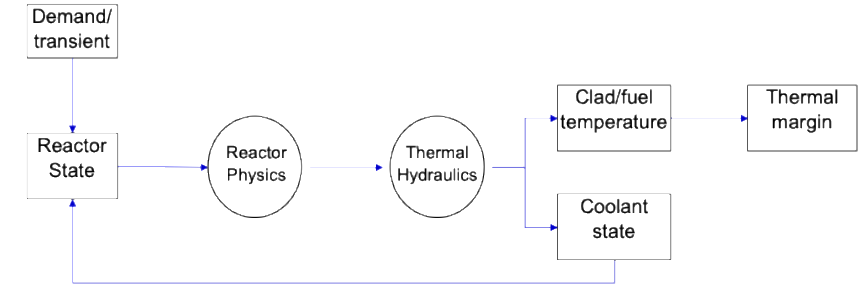
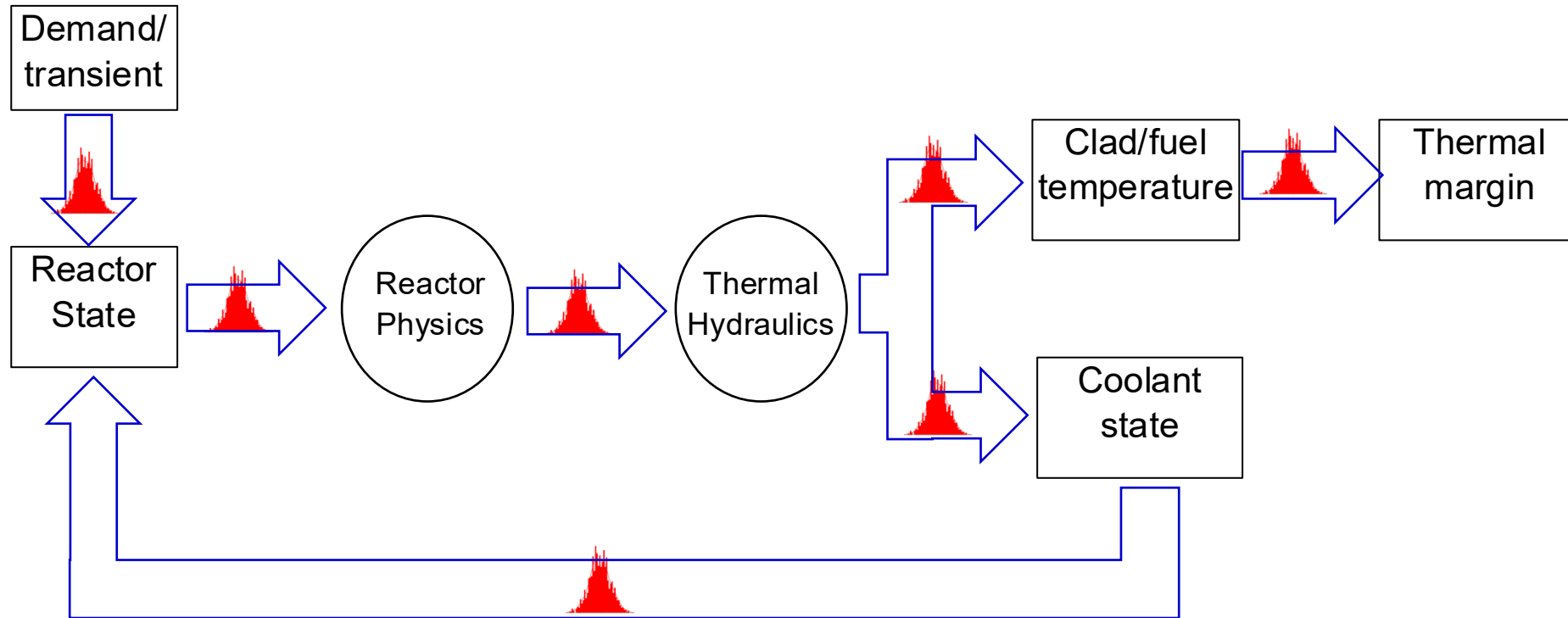
- Consider 'worst' cases (how?)



Likely to underpredict 'real' thermal margins, limiting performance

Uncertainty aware workflow

- Consider uncertainties
 - epistemic and aleatoric

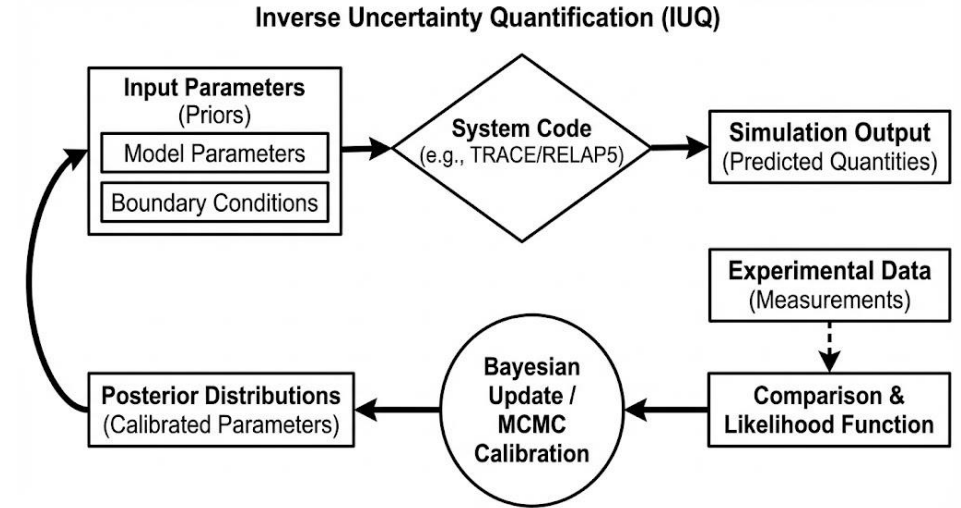


Better prediction of 'real' thermal margins, reducing conservatism

Sampling requires many point-wise predictions... hence need 'fast' models

Reactor analysis workflow

- Safety Analysis
 - Conservative/evaluation models
 - Best Estimate Plus Uncertainty (BEPU)
 - Inverse uncertainty quantification (IUQ)
- VVUQ
 - Verification
 - Establish that the code/method does what it is intended to do
 - Validation:
 - The model accurately (to some degree) reproduces experimental data
 - UQ
 - How accurate?
 - Establish error margins, sensitivities & output bounds

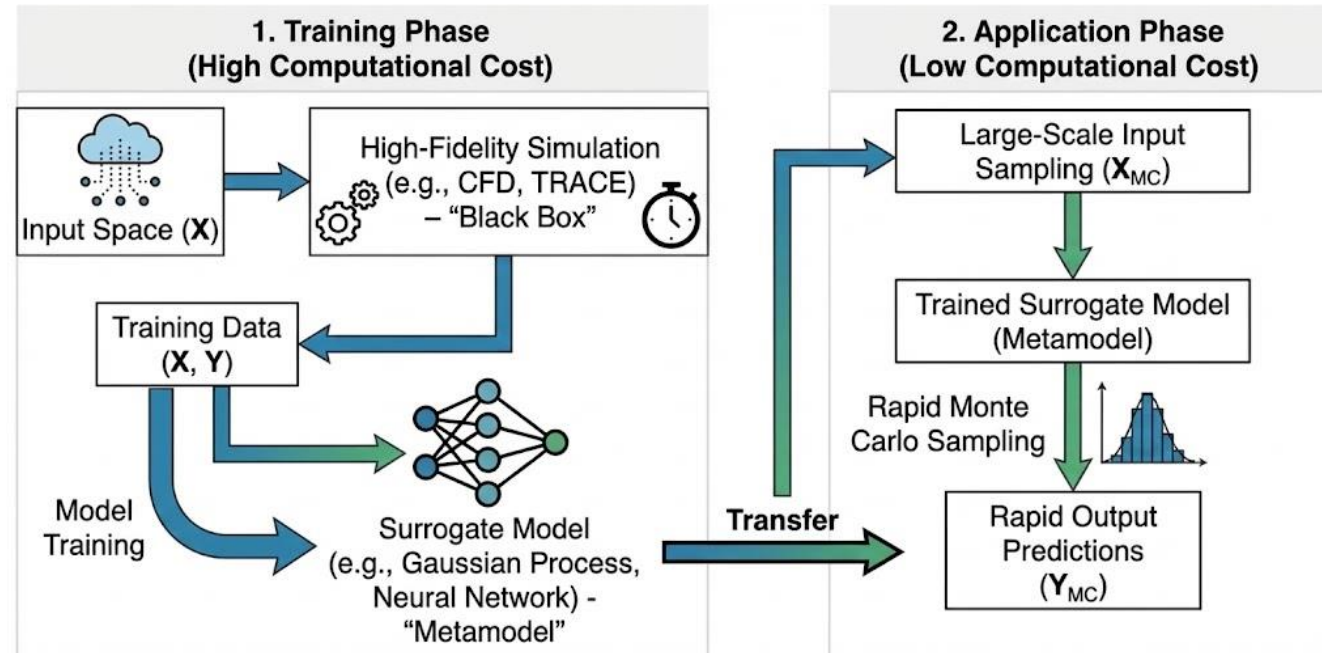


Fast models

- MOR (Physics based)
 - Simplified physics (eg LES CFD, RANS CFD)
- Surrogates (Data-driven)
 - Look-up tables (LUT) (eg Groeneveld)
 - Correlation-based (eg Dittus-Boelter, etc)
 - Polynomial response surface
 - Polynomial chaos expansion
 - Proper orthogonal decomposition (intrusive & non-intrusive)
 - Kriging (Gaussian Process Regression) – note, provides UQ directly
 - **AI/ML**
- Hybrid MOR/surrogates
 - Systems/subchannel codes use simplified physics, LUTs & correlation-based surrogates

AI/ML applications to NTH – Fast surrogates

- Data driven models
 - Random Forest, Gradient boost,...
 - Gaussian process regression
 - Neural networks
 - Many, many more
- Physics guided ML (PGML)
 - Physics informed neural networks (PINN)
 - Enforces conservation laws/PDEs as loss functions
 - Constrained architectures
 - Physical priors, kernel functions (eg in GPR)
 - Physics encoded NN (eg POD encoding)
- Generalizability
- Interpretability

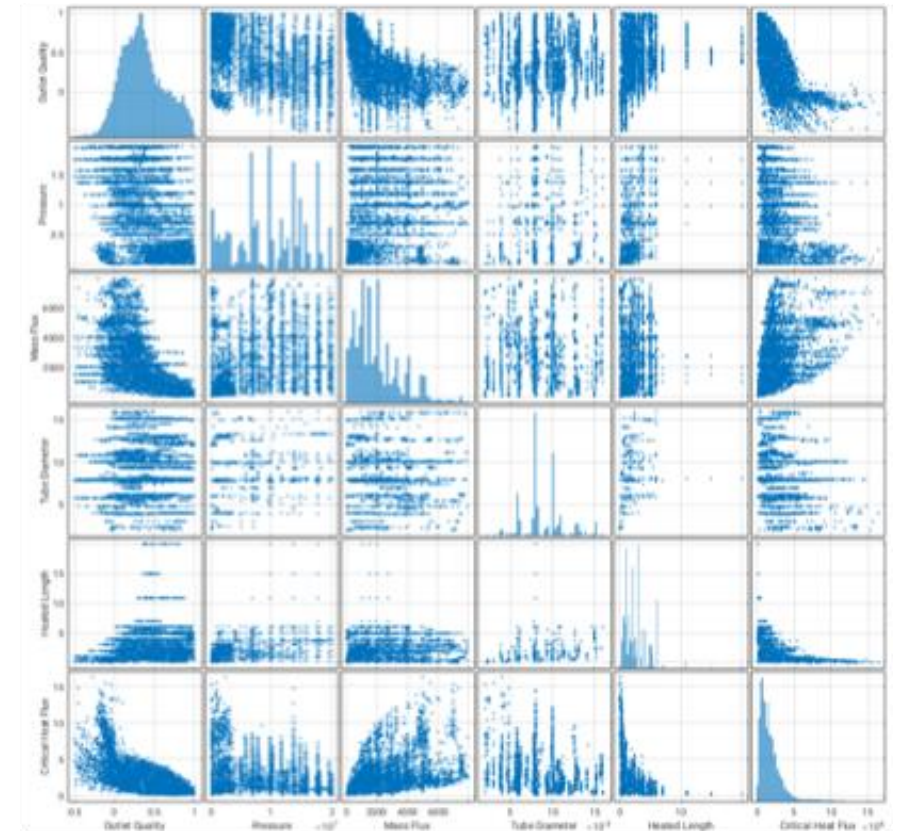


AI/ML applications to NTH - Generalization

Example: Critical Heat Flux

- Data: The Groeneveld Look-up table
 - A curated historical data-set (~30,000 points)
 - Inputs: Inlet quality, temperature, mass flowrate
 - Outputs: Critical heat flux
 - Uniform axial heating; circular pipe flow
- OECD/NEA AI/ML CHF Benchmark; Phase 1
 - ~50 submissions
 - Everything from decision trees, random forests, NN, GPR,...
 - Good 'in-data' results; great variability 'out-of data'
- OECD/NEA AI/ML CHF Benchmark; Phase 2
 - Generalizability: Non-uniform axial heating; bundle flows?
 - Uncertainty quantification?

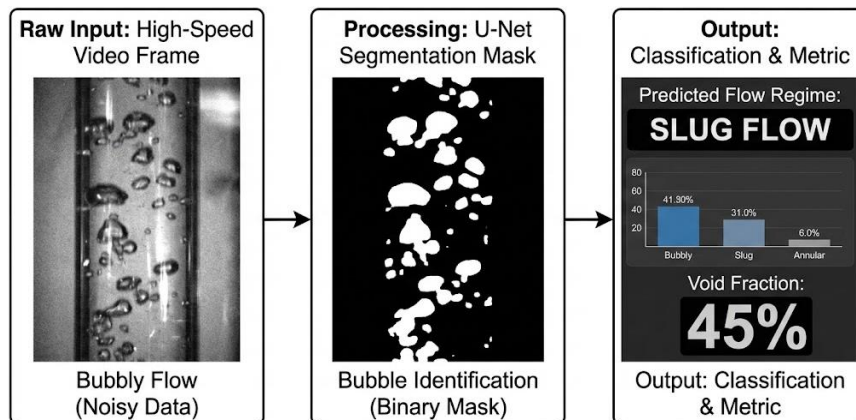
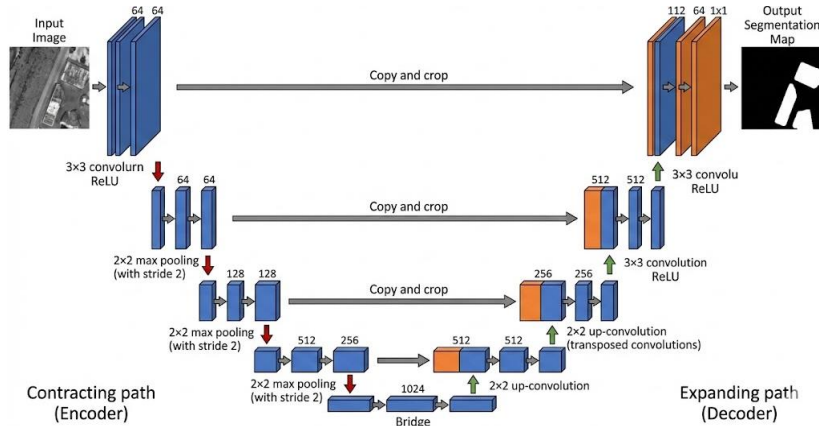
Figure 2.1. Scatter plot matrix of the NRC CHF database showing the relationship between pair of variables



- Can AI/ML *learn* physics from the GLUT and generalize?

What can AI/ML do for NTH? – Feature identification

- Feature identification



Schedule

- Day 1:
 - NTH problem framing, supervised ML, and workflow of ML
 - Lab: Baselines, preprocessing pipelines, and first ML baseline model.
- Day 2:
 - Tree ensembles for engineering tabular data.
 - Lab: Tree ensembles with regime-wise diagnostics.
- Day 3:
 - Surrogate modelling and Gaussian Processes
 - Lab: GP surrogate and extrapolation stress test.
- Day 4:
 - Neural networks for tabular engineering data, limits, and lightweight physics-guided guardrails
 - Lab: basic feed-forward neural network, guardrails, and edge-case testing.
- Day 5:
 - Practical VVUQ, and safe optimisation for decision support.

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Thank you

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